

Q2(i)

A: 2(i)

Classification into 3 domains is based on the difference of the nucleotides in the ~~BP~~ rRNA (ribosomal Ribonucleic acid) of the cell, the cell's membrane lipid structure and its sensitivity to antibiotics.

Q: 2(ii)

A: 2(ii)

A: Nucleus

B: Chloroplast

C: Vacuole

D: Cell wall

Q: 3(iii)

A: 3(ii)

Haematopoietic stem cell:-

is the type of ^{stem} cells that give rise to blood-related cell type. Any four types:-

① Red blood cells

② White blood cells

③ Platelets.

④ Nerve cells.

Q: 2(iv)

A: 2(iv)

① Place stem piece of onion skin in a drop of water on a slide and cover with a slip.

② Observe under the microscope, first under

- the low power objective.
⑤ Then observe under high power objective and draw the diagram.

Q: 2(v)

A: 2(v)

Human somatic cells are diploid = 46 chromosomes, Gametes (sperm, egg) are haploid = 23 chromosomes. Meiosis halves the chromosomes number so that fertilization ($23+23$) restores 46 in the zygote, if gametes were formed by mitosis instead, they would carry 46 chromosomes each, and the number would double every generation.

Q: 2(vi)

A: 2(vi)

Organs :- bones, cartilage, joints, tendons and ligaments.

Role in homeostasis:-

Regulates the level of calcium and other minerals in the blood, by storing and releasing them from bones, also supports/protects the body and produces blood cells.

Q:2(vii)

A:2(vii)

Fredrick Miescher (swiss physician) isolated a whitish material from the nucleus of human pus cells and named it nuclein. Based on its acidic properties. It is later renamed as nucleic acid.

Q:2(viii)

A:2(viii)

RNA is single stranded. Its pentose sugar is ribose, the base that replaces thymine is uracil. The nitrogen bases are described as overhung.

Q:2(1n)

A:2(1n)

Pepsin works best in medium acidic in stomach at 2.00 pH and trypsin works best in medium alkaline in small intestine at 8.00 pH.

Q: 2(n)

A: 2(n)

Complete oxidation of 1 glucose molecule
net 36 ATP molecules. An average human cell
uses 100 to 150 of ATP per day.

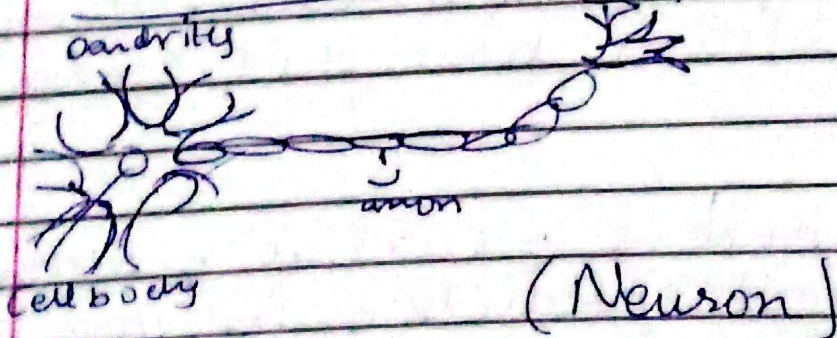
Q: 2(ni)(or)

A: 2(ni)(or)

Chewing food for longer produces
more starch digesting enzymes amylase
which mixes into the food and helps
breaks down starch. ~~Chewing~~

Q: 3(i)

A: 3(i)



Cell body: Contains the nucleus, controls
the neurons activities.
Dendrites - Conduct nerves impulse
to the cell body.

Axon: Conducts messages AWAY from the cell body.

Q:3 (ii) (or)

A:3 (i) (or)

Skeletal :-

Bones, cartilage, tendons, joints and ligaments. Regulates calcium or minerals level in blood.

Urinary :

Kidneys, ureters, bladder, urethra. Regulates water, salt, H^+ , water pH.

Lymphatic :-

Lymph, lymph nodes, lymph vessels. Maintains tissue fluid homeostasis.

Muscular :-

Skeletal, cardiac, smooth muscles. Heat production for temperature regulation.

Q. 3(iii)

A: 3(iii)

Cytosine :- 34% since C=G always pair equally, guanine :- 34%. Remaining bases (A+T) :- $100 - 68 = 32\%$ since ~~A=T~~ A=T always, adenine = 16% and thymine = 16%. each, this follows Chargaff's / Watson-Crick base pairing rule, adenine always pairs with thymine (2 hydrogen bonds), and cytosine always equals guanine (3 hydrogen bonds).

Q: 3(iv)

A: 3(iv)

Temperature:

Rate increases with temperature up to the optimum ($36^{\circ} - 38^{\circ}$ for human enzymes) then decreases sharply due to denaturation.

pH :-

Each enzymes work best at specific pH (called optimum pH). for example (trypsin and pepsin) are both protein digesting enzyme. Pepsin works best at ~~stomach~~ acidic stomach at 2.00 pH and trypsin works best in medium.

alkaline in small intestine at 8.00 pH.

Substrate Concentration:-

Rate increases proportionally with substrate concentration then plateaus once all active sites are occupied called saturation of all active site.

M.C.Qs:-

1. C

2. B

3. C

4. D

5. C

6. D

7. B

8. C

9. C

10. D

11. B

12. E